

IAM User and AWS Configuration

The screenshot shows the AWS IAM console interface. On the left is a navigation menu with sections like 'Identity and Access Management (IAM)', 'Access Management', and 'Access reports'. The main content area is titled 'terraform-user-2' and includes a 'Delete' button. It features a 'Summary' section with fields for ARN, Console access status, Access key 1, Access key 2, Created date, and Last console sign-in. Below this are tabs for 'Permissions', 'Groups', 'Tags', 'Security credentials', and 'Last Accessed'. The 'Security credentials' tab is active, showing 'Console sign-in' with a link and 'Console password' status. It also displays 'Multi-factor authentication (MFA)' options with buttons to 'Remove', 'Resync', or 'Assign MFA device'. A table below the MFA section shows no devices are currently assigned.

The screenshot shows a terminal window with a dark theme. The left sidebar shows a file explorer with 'TERRAFORM-EC2' selected. The terminal output shows the following commands and results:

```
richie@richie-Latitude-5490:~/Azubi Africa/terraform-ec2$ aws configure
AWS Access Key ID [*****]: AKIAIYKJ5ZY4P6YT2LWC
AWS Secret Access Key [*****]: 2880UC30MDNV03Pp5H1toIfexK01WYJKVbAMM01
Default region name [us-east-2]:
Default output format [json]: json

richie@richie-Latitude-5490:~/Azubi Africa/terraform-ec2$ aws s3 ls

An error occurred (AccessDenied) when calling the ListBuckets operation: User: arn:aws:iam::533267140152:user/terraform-user-2 is not authorized to perform: s3:ListAllMyBuckets because no identity-based policy allows the s3:ListAllMyBuckets action

richie@richie-Latitude-5490:~/Azubi Africa/terraform-ec2$ aws ec2 ls

aws: [ERROR]: argument operation: Found invalid choice 'ls'

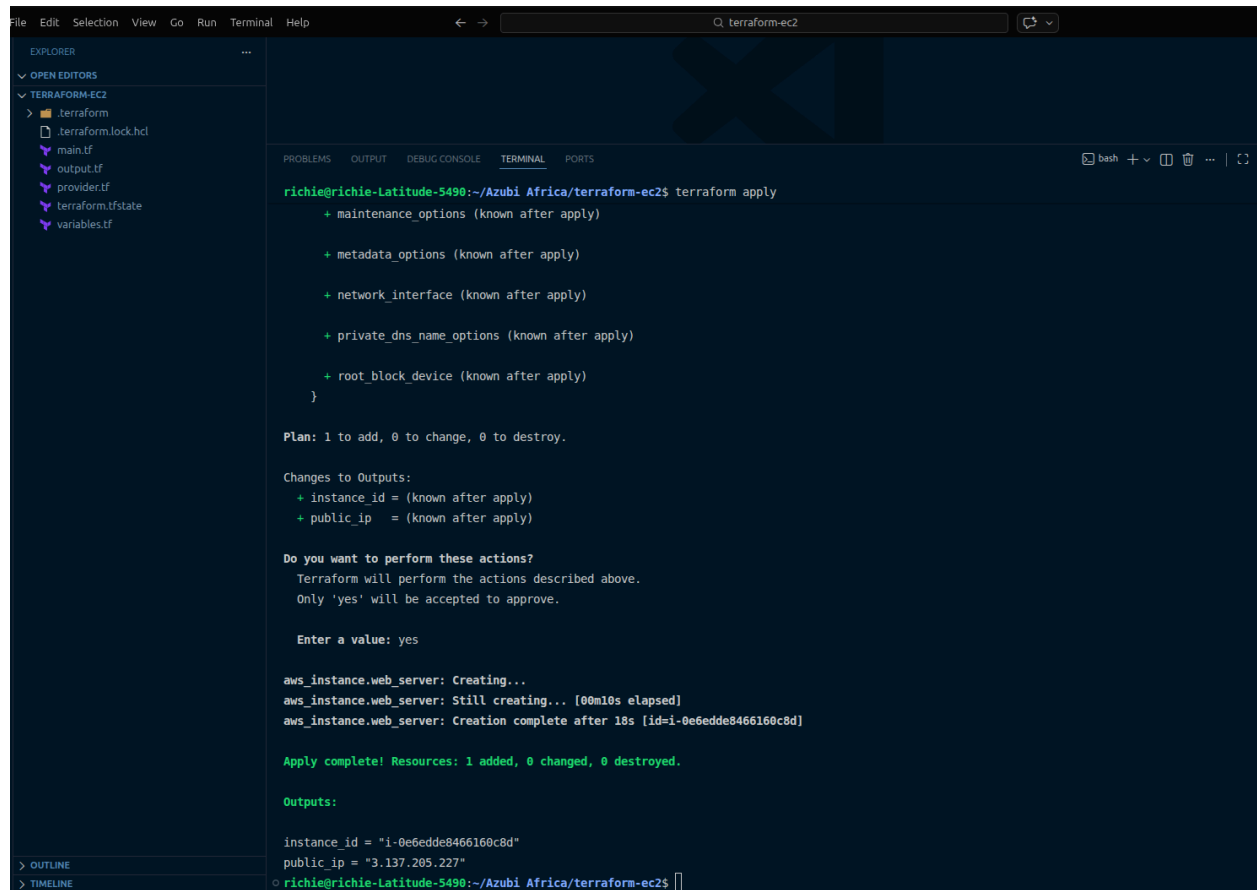
usage: aws [options] <command> <subcommand> [-<subcommand> ...] [parameters]
To see help text, you can run:

    aws help
    aws <command> help
    aws <command> <subcommand> help

richie@richie-Latitude-5490:~/Azubi Africa/terraform-ec2$ aws ec2 describe-instances
{
  "Reservations": []
}

richie@richie-Latitude-5490:~/Azubi Africa/terraform-ec2$
```

Terraform project setup, initialization and apply configuration

A screenshot of a Visual Studio Code editor window. The Explorer sidebar on the left shows a project named 'TERRAFORM-EC2' with files: '.terraform.lock.hcl', 'main.tf', 'output.tf', 'provider.tf', 'terraform.tfstate', and 'variables.tf'. The main editor area displays the terminal output of a 'terraform apply' command. The output shows a plan for adding resources, a confirmation prompt, and the successful creation of an AWS EC2 instance. The terminal text is as follows:

```
richie@richie-Latitude-5490:~/Azubi Africa/terraform-ec2$ terraform apply
+ maintenance_options (known after apply)

+ metadata_options (known after apply)

+ network_interface (known after apply)

+ private_dns_name_options (known after apply)

+ root_block_device (known after apply)
}

Plan: 1 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ instance_id = (known after apply)
+ public_ip   = (known after apply)

Do you want to perform these actions?
Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

aws_instance.web_server: Creating...
aws_instance.web_server: Still creating... [00m10s elapsed]
aws_instance.web_server: Creation complete after 18s [id=i-0e6edde8466160c8d]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Outputs:

instance_id = "i-0e6edde8466160c8d"
public_ip   = "3.137.205.227"
richie@richie-Latitude-5490:~/Azubi Africa/terraform-ec2$
```

EC2 instance provisioned

EC2 > Instances

Instances (1/1) Info

Last updated less than a minute ago

Connect Instance state Actions Launch instances

Instance state: running

Find Instance by attribute or tag (case-sensitive)

Clear filters

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP
TerraformDem...	i-0e6edde8466160c8d	running	t3.micro	3/3 checks passed	us-east-2c	ec2-3-137-205-227.us-...	3.137.205.227	-

i-0e6edde8466160c8d (TerraformDemoServer)

Details Status and alarms Monitoring Security Networking Storage Tags

Instance summary

Instance ID
i-0e6edde8466160c8d

IPv6 address
-

Hostname type
IP name: ip-172-31-36-18.us-east-2.compute.internal

Answer private resource DNS name
-

Public IPv4 address
3.137.205.227 | open address

Instance state
Running

Private IP DNS name (IPv4 only)
ip-172-31-36-18.us-east-2.compute.internal

Instance type
t3.micro

Private IPv4 addresses
172.31.36.18

Public DNS
ec2-3-137-205-227.us-east-2.compute.amazonaws.com | open address

Elastic IP addresses
-

CloudShell Feedback Console Mobile App

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Added User data

main.tf

Terraform-EC2

terraform

terraform.lock.hcl

main.tf

output.tf

provider.tf

terraform.tfstate

terraform.tfstate.backup

variables.tf

```
1 data "aws_ami" "amazon_linux" {
2
3 }
4
5
6
7
8
9
10 resource "aws_instance" "web_server" {
11
12   ami           = data.aws_ami.amazon_linux.id
13   instance_type = var.instance_type
14
15   user_data = <<-EOF
16     #!/bin/bash
17     yum update -y
18     yum install httpd -y
19     systemctl start httpd
20     systemctl enable httpd
21
22     echo "<html>
23     <head><title>Terraform Web Server</title></head>
24     <body>
25     <h1>Hello from Terraform EC2!</h1>
26     <p>Apache Server Installed Successfully</p>
27     <h2 style='background-color: #293049; color: #f0ad0d;'>Welcome to our group</h2>
28   </EOF>
29
30
31
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

richie@richie-Latitude-5490:~/Azubi Africa/terraform-ec2\$ terraform apply

+ user_data = "442bb9ce6135d41450b96d8a73734a35f72793ff"

(37 unchanged attributes hidden)

(8 unchanged blocks hidden)

Plan: 0 to add, 1 to change, 0 to destroy.

Changes to Outputs:

- public_ip = "3.137.205.227" -> (known after apply)

Prepare interview with project. Completed

Integrate Flutterwave payment flow. Completed

Implement combined signup/login page. Completed

MORE (18)

main.tf

Describe what to build next